



Shri Vile Parle Kelavani Mandal's  
**INSTITUTE OF TECHNOLOGY**  
**DHULE (M.S.)**  
**DEPARTMENT OF COMPUTER ENGINEERING**

|                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-----------------------|
| <b>Subject :</b> Machine Learning Lab                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Remark</b>        |                       |
| <b>Name:</b> _____ <b>Roll No. :</b> _____                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |                       |
| <b>Class:</b> T.Y. Comp Engg. <b>Batch :</b> T <b>Division:</b> _____                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Signature</b>     |                       |
| <b>Expt. No. :</b> 9 <b>Date :</b> /    /                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |                       |
| <b>Title :</b> Write a program to implement the Hierarchical Agglomerative Clustering Algorithm for a sample training data set stored as a .CSV file. Compute the accuracy of the classifier, considering few test data sets and Visualize model with Python. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |                       |
| <b>Rubrics:</b>                                                                                                                                                                                                                                               | <b>Date of Submission:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                      |                       |
|                                                                                                                                                                                                                                                               | <b>Marks Split up to</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Maximum Marks</b> | <b>Marks Obtained</b> |
|                                                                                                                                                                                                                                                               | <b>Performance/conduction</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 3                    |                       |
|                                                                                                                                                                                                                                                               | <b>Report Writing</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 3                    |                       |
|                                                                                                                                                                                                                                                               | <b>Attendance</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2                    |                       |
|                                                                                                                                                                                                                                                               | <b>Viva/Oral</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2                    |                       |
|                                                                                                                                                                                                                                                               | <b>Total Marks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 10                   |                       |
|                                                                                                                                                                                                                                                               | <b>Aim :</b><br><br>The main aim of this lab is to demonstrate the how to Hierarchical Agglomerative Clustering Algorithm with Python<br><br>Secondary aim of this lab is to understand <ul style="list-style-type: none"><li>What is Hierarchical Agglomerative Clustering Algorithm?</li><li>How Hierarchical Agglomerative Clustering works?</li><li>What is Dendrograms?</li><li>What is Linkage Criteria?</li><li>How to Fit a Hierarchical Agglomerative Clustering using Scikit-Learn?</li><li>How to Visualize Hierarchical Agglomerative Clustering using Matplotlib?</li></ul> |                      |                       |
| <b>Software Required :</b><br>Jupyter Notebook , Python 3.6, Or Google Colab                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |                       |
| <b>Dataset</b>                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |                       |

**Used**

Mall\_cusomer.csv

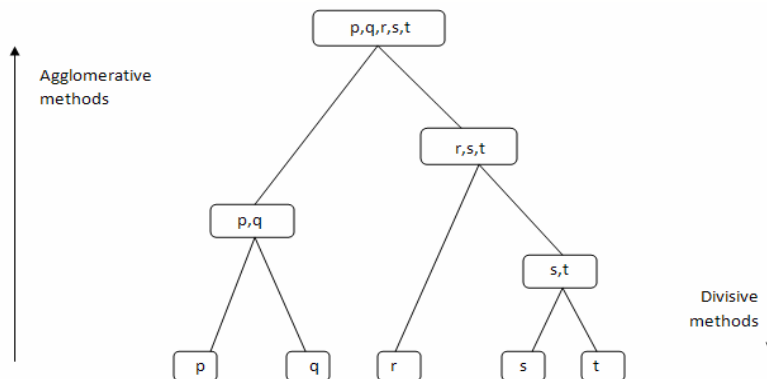
**Theory :**

## Introduction to Hierarchical Clustering

Hierarchical clustering algorithms group similar objects into groups called clusters. There are two types of hierarchical clustering algorithms:

**Agglomerative** — Bottom up approach. Start with many small clusters and merge them together to create bigger clusters.

**Divisive** — Top down approach. Start with a single cluster than break it up into smaller clusters.



## Some pros and cons of Hierarchical Clustering

### Pros

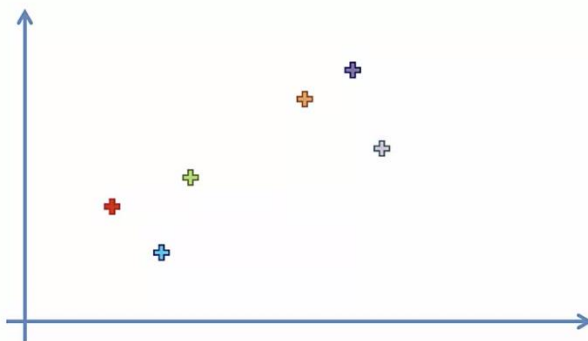
- No assumption of a particular number of clusters (i.e. k-means)
- May correspond to meaningful taxonomies

### Cons

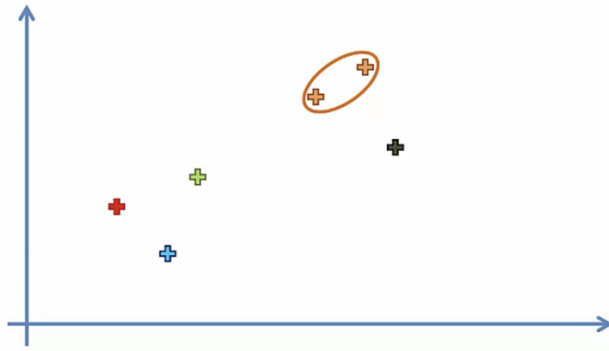
- Once a decision is made to combine two clusters, it can't be undone
- Too slow for large data sets,  $O(n^2 \log(n))$

### How it works

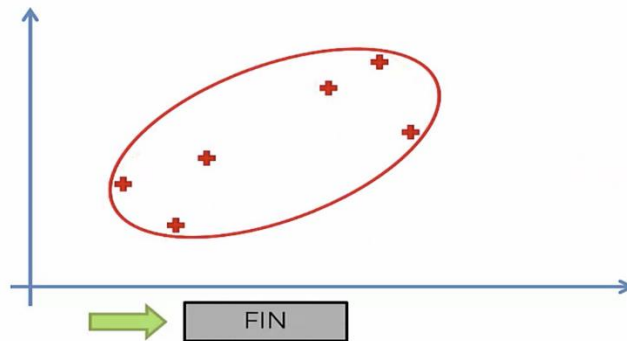
1. Make each data point a cluster



2. Take the two closest clusters and make them one cluster



3. Repeat step 2 until there is only one cluster

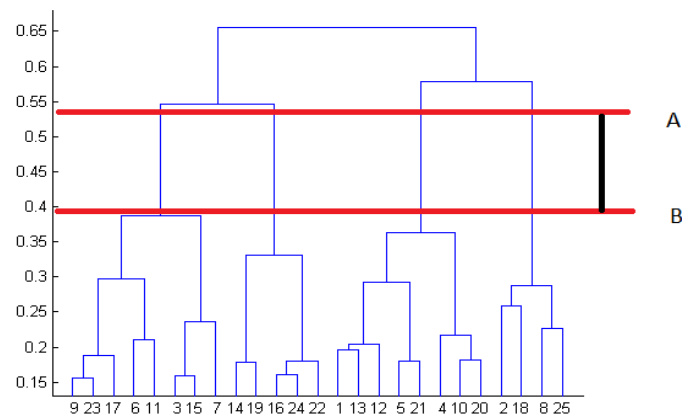


### Dendrograms

We can use a dendrogram to visualize the history of groupings and figure out the optimal number of clusters.

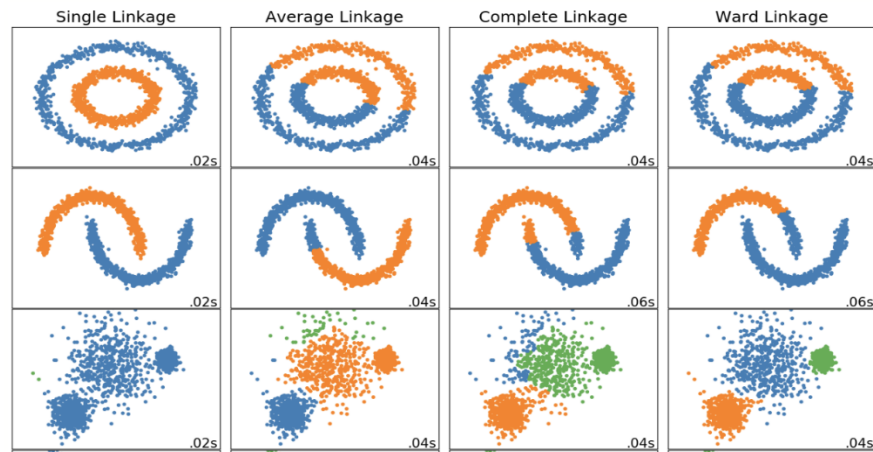
1. **Determine the largest vertical distance that doesn't intersect any of the other clusters**
2. Draw a horizontal line at both extremities
3. The optimal number of clusters is equal to the number of vertical lines going through the horizontal line

For eg., in the below case, best choice for no. of clusters will be 4.



## Linkage Criteria

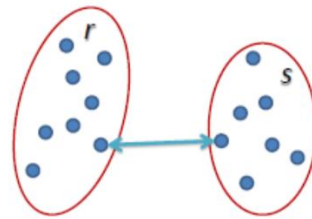
Similar to gradient descent, you can tweak certain parameters to get drastically different results.



The linkage criteria refers to how the distance between clusters is calculated.

### Single Linkage

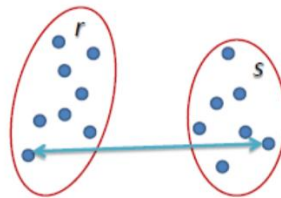
The distance between two clusters is the shortest distance between two points in each cluster



$$L(r, s) = \min(D(x_{ri}, x_{sj}))$$

### Complete Linkage

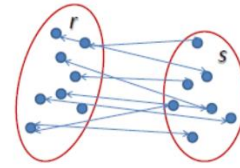
The distance between two clusters is the longest distance between two points in each cluster



$$L(r, s) = \max(D(x_{ri}, x_{sj}))$$

### Average Linkage

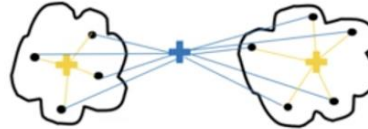
The distance between clusters is the average distance between each point in one cluster to every point in other cluster



$$L(r, s) = \frac{1}{n_r n_s} \sum_{i=1}^{n_r} \sum_{j=1}^{n_s} D(x_{ri}, x_{sj})$$

### Ward Linkage

The distance between clusters is the sum of squared differences within all clusters

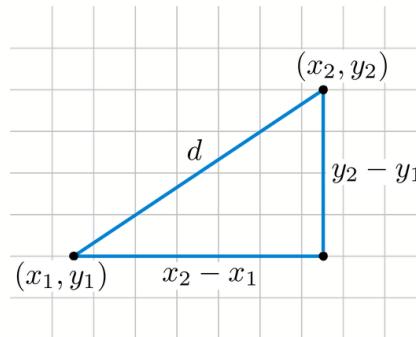


### Distance Metric

The method you use to calculate the distance between data points will affect the end result.

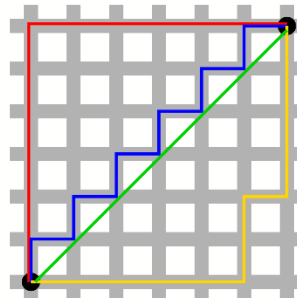
### Euclidean Distance

The shortest distance between two points. For example, if  $x=(a,b)$  and  $y=(c,d)$ , the Euclidean distance between  $x$  and  $y$  is  $\sqrt{(a-c)^2+(b-d)^2}$



### Manhattan Distance

Imagine you were in the downtown center of a big city and you wanted to get from point A to point B. You wouldn't be able to cut across buildings, rather you'd have to make your way by walking along the various streets. For example, if  $x=(a,b)$  and  $y=(c,d)$ , the Manhattan distance between  $x$  and  $y$  is  $|a-c|+|b-d|$



|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Procedure</b>          | <ol style="list-style-type: none"> <li>1. Import the required libraries.</li> <li>2. Import the Dataset.</li> <li>3. Using the dendrogram to find the optimal number of clusters<br/><br/>Looking at the dendrogram, the highest vertical distance that doesn't intersect with any clusters is the middle green one. Given that 5 vertical lines cross the threshold, the optimal number of clusters is 5.</li> <li>4. Training the Hierarchical Clustering model on the dataset<br/><br/>Create an instance of AgglomerativeClustering using the euclidean distance as the measure of distance between points and ward linkage to calculate the proximity of clusters.</li> <li>5. Creating Output labels for Generating Graph</li> <li>6. Visualize the clusters</li> </ol> |
| <b>Practice Problems:</b> | <ol style="list-style-type: none"> <li>1. Implement Hierarchical Agglomerative Clustering Algorithm</li> <li>2. Visualize your build model in python</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Conclusion</b>         | <p>In this way we successfully implement and visualize Hierarchical Agglomerative Clustering Algorithm in python.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

**Test your  
Knowledge**

1. For the model we created in lab session write the following results

|                                             |  |
|---------------------------------------------|--|
| Optimal cluster Number                      |  |
| Cluster Leaves                              |  |
| Rand Score                                  |  |
| number of connected components in the graph |  |
| Number of features seen during fit          |  |

2. Create the model using following parameter setting and write down the result

**Linkage = Single**

**Metric = cosine**

|                                             |  |
|---------------------------------------------|--|
| Optimal cluster Number                      |  |
| Cluster Leaves                              |  |
| Rand Score                                  |  |
| number of connected components in the graph |  |
| Number of features seen during fit          |  |